

Haris Tahir Rana

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Education

Lahore University of Management Sciences

BSc (Hons.) in Computer Science (Artificial Intelligence Specialization)

Cumulative GPA 3.37/4.0

Relevant Courses: Data Mining, Data Science, Databases, Deep Learning, Generative AI, NLP, Software Engineering

Lahore, Punjab
May 2022 - July 2026

Lahore Grammar School

A-Level - 3 A* 1 B; SAT Score 1510

Lahore, Punjab
May 2022

Experience

CLAPP HOSPITAL

Machine Learning Engineer

- Developed an automated diagnostic system to classify hypernasal vs. normal speech patterns, implementing a speaker diarization pipeline to isolate 5,349 pediatric speech segments from 352 minutes of raw clinic audio and eliminate adult assessor voice interference.
- Engineered a 45-dimensional acoustic feature extraction pipeline optimized to 30 key diagnostic markers; implemented patient-level cross-validation to prevent data leakage and secure a 91.16% baseline classification accuracy.
- Selected XGBoost over deep learning architectures for interpretability and robustness on a constrained dataset, then applied majority-vote aggregation across patient segments to lift diagnostic accuracy by ~3%, achieving 96.97% patient-level accuracy and 98.08% sensitivity on a 99-patient held-out test set.

Lahore, Punjab
Feb 2026 – Apr 2026

COMPUTER VISION AND GRAPHICS LAB

Deep Learning and Medical Imaging Researcher

- Designed and implemented an end-to-end deep learning pipeline for automated TB detection, integrating UNet++ segmentation with SEResNet50 and Swin Transformer-based classification models.
- Curated, preprocessed, and trained models on 10,000+ chest X-ray images from multiple public datasets, achieving 99% classification accuracy and 96% Dice score for segmentation.
- Evaluated model performance via IoU (0.9173) and macro F1-score (0.99), applying GradCAM++ for explainability; authored an unpublished research manuscript documenting methodology and results.

Lahore, Punjab
Feb 2025 – May 2025

ROZEE.PK

Full-Stack Web Developer Intern

- Built a voice-enabled WhatsApp bot using Node.js and JavaScript that guided users through CV creation by asking questions via voice and processing multilingual voice responses.
- Used OpenAI and Azure Speech APIs for real-time voice-to-text and language translation, enabling inclusive access for users with limited literacy.
- Developed a separate chatbot-based CV builder using simple text messaging, integrated with Rozee.pk's platform via Laravel (PHP), to make resume creation intuitive and accessible for all users.

Lahore, Punjab
Jul 2024 – Aug 2024

Projects

ARTHUS SYSTEMS COMPANY MANAGEMENT SYSTEM

Technologies: React, TypeScript, Next.js, PostgreSQL

- Developed a centralized CRM and automated outreach system with a custom browser extension for lead ingestion, throttled two-way email pipeline for personalized outreach, processing 350–400 leads/week and cutting manual outreach time by over 75%.
- Engineered an automated double-entry accounting framework with vision-based receipt parsing, cutting per-receipt processing time from ~60 seconds of manual entry to a few seconds via drag-and-drop; also built an encrypted data vault to securely track milestones and credentials across multiple projects.

Jun 2026 – Jul 2026

POWER ZONE INVENTORY MANAGEMENT SYSTEM

Technologies: React, TypeScript, Next.js, PostgreSQL

- Designed and deployed an end-to-end enterprise inventory platform to replace a legacy spreadsheet system, automating workflows from factory receiving to order fulfillment and reducing inventory mismatches and tracking errors by over 90%.
- Engineered a secure transactional database engine using atomic PostgreSQL functions, row-level locking, and Row-Level Security (RLS) to guarantee data integrity and enforce strict role-based access control (RBAC) across 8 distinct modules.

Apr 2026 – Jul 2026

AGENTIC 10-K INTELLIGENCE SYSTEM

Technologies: LangChain, ChromaDB, Google AI Studio

- Built an automated Q&A and document ingestion pipeline for SEC 10-K filings, implementing semantic partitioning and HTML-to-Markdown table parsing to maintain financial data integrity and boosting RAGAS Faithfulness from 0.18 to 0.74 using various prompting techniques.
- Engineered an autonomous Plan-and-Execute research agent to track trends across 50 multi-year filings, integrating a custom Python-based calculator tool for precise ratio analysis and a summarization engine that delivered investment memos with a 4.39/5.0 G-Eval coherence score.

Oct 2025 – Dec 2025

HEALTHSYNC

Technologies: React Native, TypeScript, Django, MongoDB

- Led a 5-member team to design and build a health mobile app prototype connecting patients directly with verified doctors via live chat and automated appointment scheduling.
- Engineered a secure provider onboarding workflow using Firebase Storage for credential verification, with an integrated AI chatbot for patient support queries.

Jan 2025 – May 2025

Leadership and Activities

FINTRA-LUMS

Researcher

- Authored and published "How Can the Ehsaas Program Be a Financial Revolution for the Masses of Pakistan?" in *Bottomline*, FINTRA-LUMS's annual finance magazine, analyzing the program's economic impact, financial inclusion mechanisms, and effectiveness in poverty alleviation.

Lahore, Punjab
Sep 2023 – May 2024

LAHORE GRAMMAR SCHOOL

Student Council Prefect

- Organized and supervised major school events, including Society Fairs, Inter-School Competitions, Sports Fests, and Parent-Teacher conferences, engaging the entire 200+ student body.

Lahore, Punjab
Sep 2021 – May 2022

Skills & Interests

Technical: Python, TypeScript, React, React Native, Next.js, LangChain, Google AI Studio, MongoDB, Git, Numpy, Pandas, Power BI, SQL, Stata, Pytorch, Scikit-Learn

Certifications: Power BI (Datacamp); Data Analytics Essentials (Cisco); AI Fundamentals (IBM)

Language: Fluent in English, Urdu; Limited Proficiency in Arabic, Turkish

Interests: Tennis, Travelling, Geopolitics, Poetry